Polynomial System Solving via Factor Branching

Ilay Menahem[a], Michael Shalyt[a] and Ido Kaminer[a,b]

[a]Andrew and Erna Viterbi Faculty of Electrical and Computer Engineering, Technion – Israel Institute of Technology, Haifa, 3200003, Israel
[b]Department of Materials Science and Engineering, Technion – Israel Institute of Technology, Haifa, 3200003, Israel

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ABSTRACT

Polynomial systems are a fundamental language for describing algebraic and geometric problems in mathematics and its applications. Beyond finding individual solutions, one often seeks global properties of the solution space, such as its dimension and irreducible components, or structural properties and decompositions of the underlying polynomial ideal. For large polynomial systems with many variables and high degrees, determining these global properties can become intractable. Leading approaches include variants of Buchberger's algorithm, and resultants. However, even the most modern approaches still often fail on practical problems or slight perturbations of standard benchmarks. Here we introduce a framework that divides a polynomial system into simpler branches by interreducing and factoring its equations. The framework only assumes that the solver used on each branch describes satisfiable (SAT) branches and certifies unsatisfiable (UNSAT) ones. In our experiments, we use Gröbner-basis solvers and apply the method to a system arising from research in quantum field theory for which no standard method in our comparison succeeds. Together, these ideas offer a path toward the exact analysis of previously inaccessible polynomial systems.

1. Introduction

Systems of multivariate polynomial equations appear whenever a symbolic model contains nonlinear algebraic constraints. Applications include robotics, coding theory, optimization, statistics, and physics (Sturmfels, 2002). In many of these problems, an isolated solution is not enough. One needs the dimension of the solution set, its irreducible components, or the exact relations that hold on every solution. These are questions about the ideal I generated by the equations and its variety $V(I)$, rather than about a single point (Cox et al., 2015; Kreuzer and Robbiano, 2000).

The outputs of algebraic algorithms vary along two axes (Figure 1):

- How far the output breaks down the ideal: it treats I as one piece, decomposes it into ideals that need not be prime or primary, or decomposes it into its prime or primary components.
- How much information the output retains: whether $V(I)$ is empty, properties of $V(I)$ such as its dimension, $\sqrt{I}$, or I itself.

Elimination by resultants provided an early systematic way to attack polynomial systems, but the introduction of Gröbner bases gave a common algorithmic framework for elimination, ideal membership, dimension, and related tasks (Cox et al., 2015; Kreuzer and Robbiano, 2000). Buchberger developed this framework in his 1965 thesis (Buchberger, 2006). Later algorithms changed how the expensive reductions are organized. F4 replaces individual polynomial reductions with sparse linear algebra, while F5 uses signatures to avoid many reductions to zero (Faugère, 1999, 2002). Primary-decomposition algorithms use Gröbner bases and factorization to separate components (Gianni et al., 1988). Factorizing Gröbner-basis methods split on individual polynomial factorizations and preserve the solution set; earlier methods also interreduce and prune redundant branches (Decker et al., 2025; Gräbe, 1995). Numerical algebraic geometry takes a different route: homotopy continuation computes isolated roots numerically, while witness sets represent positive-dimensional components (Sommese et al., 2001; Sommese and Wampler, 2005).

Recent work has also explored machine-learning methods for polynomial systems. Peifer et al. trained a reinforcement-learning policy for pair selection in Buchberger's algorithm (Peifer et al., 2020). Kera et al. trained transformers to compute Gröbner bases directly for a restricted class of zero-dimensional radical ideals (Kera et al., 2024). Yet these works are limited to small and specific polynomial ideals.

Despite this progress, large systems such as a system that came up in our research of quantum gravity can remain inaccessible.

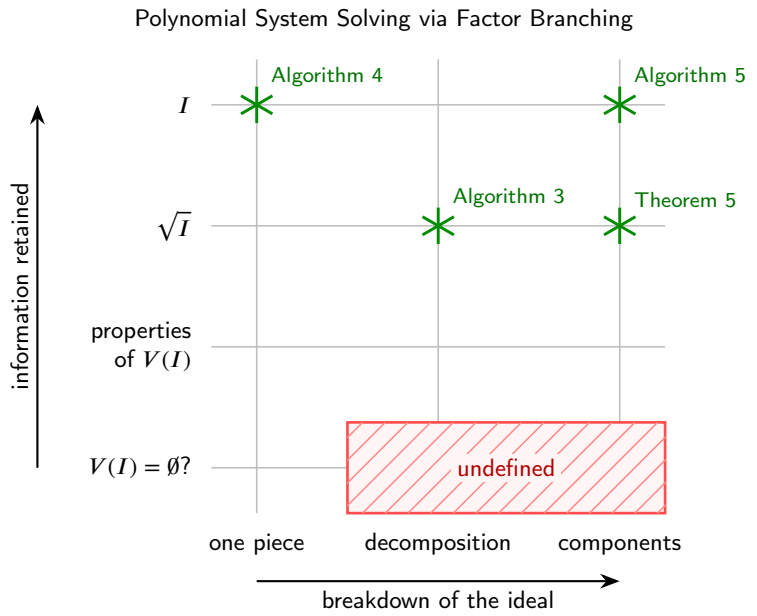


Figure 1: Outputs of polynomial-system algorithms on two axes. Columns order the outputs by how far they break down the ideal I, and rows by how much information they retain. Crosses mark the methods of this paper. The hatched cells are undefined, since whether $V(I)$ is empty is a single answer with nothing to break down.

Here, we introduce solver-agnostic methods for several of these outputs, all built from the same steps. Each method factors the equations and enumerates the minimal hitting sets of the resulting factor sets. A minimal hitting set contains at least one factor of every equation, so it defines a branch, a smaller system whose solutions are solutions of the original one, and $V(I)$ is the union of their solution sets. Interreducing and factoring a branch can split it further, and the methods repeat the split until it changes nothing. Each method then calls an exact solver on the branches and combines the results into its output. Since the worst-case complexity of computing a Gröbner basis is doubly exponential in the number of variables (Mayr and Meyer, 1982), breaking down a large system into smaller branches can be substantially faster than solving the original system directly.

Figure 1 places the methods of this paper on the two axes. The methods differ in the branch solver and in how they combine the branches. Factor branching asks the solver to describe each branch, including any positive-dimensional families, or to certify that it is UNSAT, and so returns a decomposition of $V(I)$ whose pieces may overlap (Section 3). Prime factor branching asks for the minimal primes of each branch and returns those of I, which give the irreducible components of $V(I)$ over K and their dimensions (Theorem 5). The remaining methods keep the exponents of the factors, and with them I itself. Ideal reduction uses the SAT branches to replace generators by smaller members of I (Subsection 4.1). Primary factor branching obtains the isolated primary components of I from its minimal primes and runs a general primary decomposition only when I has embedded components (Subsection 4.2).

We prove that every method is exact. Refinement terminates and preserves the solution set, ideal reduction preserves I, and prime and primary factor branching return the minimal primes and a minimal primary decomposition of I. Every numbered result is also formalized in Lean 4 with Mathlib, in the repository listed under Data availability. On generated systems with factor structure and on size-graded families from SymbolicData and from applications, factor branching with the best of its backends completes 19 of 20 and 27 of 31 systems within 60 seconds, against at most 16 and 20 for the direct Gröbner-basis and decomposition methods. On a system from quantum field theory, all nine standalone baselines time out, whereas factor branching with the `facstd` backend completes in under a second. Primary factor branching computes the primary decomposition of this system in under four minutes, while Singular's `primdecSY` does not finish within 30 minutes on the original equations.

2. Preliminaries and notation

2.1. Polynomial systems, ideals, and varieties

Let K be a field, let $\overline{K}$ be its algebraic closure, and write $R = K[x_1, \dots, x_n]$. The input is a finite system $P = \{p_1, \dots, p_m\} \subseteq R$ of nonzero polynomials. Unless stated otherwise, a solution is a point in $\overline{K}^n$. For a finite set $Q \subseteq R$, its solution set is

$$V(Q) = \{a \in \overline{K}^n : q(a) = 0 \text{ for every } q \in Q\}.$$

We call Q satisfiable (SAT) if $V(Q) \neq \emptyset$ and unsatisfiable (UNSAT) if $V(Q) = \emptyset$.

We write $\langle Q \rangle$ for the ideal generated by Q.

For $X \subseteq \overline{K}^n$, its vanishing ideal $I(X)$ contains every polynomial in R that vanishes on all of X. The radical of an ideal J is $\sqrt{J} = \{g \in R : g^r \in J \text{ for some } r \geq 1\}$. Hilbert's Nullstellensatz gives

$$I(V(Q)) = \sqrt{\langle Q \rangle}.$$

Thus $V(Q) = \emptyset$ exactly when $\langle Q \rangle = \langle 1 \rangle$ (Cox et al., 2015; Kreuzer and Robbiano, 2000).

2.2. Saturation and decompositions

The saturation of an ideal J by $u \in R$ (Cox et al., 2015, Ch. 4, §4) is

$$J : u^\infty = \{g \in R : u^m g \in J \text{ for some } m \geq 0\}.$$

An ideal $\mathfrak{q}$ is primary if $gh \in \mathfrak{q}$ implies $g \in \mathfrak{q}$ or $h^m \in \mathfrak{q}$ for some $m \geq 1$, and then $\sqrt{\mathfrak{q}}$ is prime. Every ideal J of R has a minimal primary decomposition (Cox et al., 2015, Ch. 4, §8)

$$J = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_t$$

into primary ideals with distinct radicals $\mathfrak{p}_j = \sqrt{\mathfrak{q}_j}$, where no $\mathfrak{q}_j$ contains the intersection of the others. The radicals are the associated primes of J, and the minimal ones among them are the minimal primes of J. The minimal primes intersect to $\sqrt{J}$ (Cox et al., 2015, Ch. 4, §6). The components whose radicals are minimal primes, called isolated, are unique; the other components, called embedded, are not (Cox et al., 2015, Ch. 4, §8). Saturation keeps the components whose radicals do not contain u, $J : u^\infty = \bigcap_{u \notin \mathfrak{p}_j} \mathfrak{q}_j$ (Atiyah and Macdonald, 1969, Prop. 4.9). A prime that contains the intersection of finitely many ideals contains one of them (Atiyah and Macdonald, 1969, Prop. 1.11).

2.3. Monomial orders and Gröbner bases

Let $\mathbb{M}_n = \{x^\alpha : \alpha \in \mathbb{N}^n\}$ be the monomials of R. A monomial order $\prec$ is a total order on $\mathbb{M}_n$ such that every nonempty subset has a least element and

$$u \prec v \implies uw \prec vw \qquad (u, v, w \in \mathbb{M}_n).$$

For a nonzero polynomial p, the leading monomial $\mathrm{LM}(p)$ is the greatest monomial of p under $\prec$, $\mathrm{LC}(p)$ is its coefficient, and $\mathrm{LT}(p)$ is its leading term.

For an ideal J, define its leading-monomial ideal by

$$\mathrm{LM}(J) = \langle \mathrm{LM}(f) : f \in J \setminus \{0\} \rangle.$$

A finite set $G \subseteq J$ is a Gröbner basis of J if

$$\mathrm{LM}(J) = \langle \mathrm{LM}(g) : g \in G \rangle.$$

A finite set Q is *interreduced* if every element is monic and no term of one element is divisible by the leading monomial of another. Algorithm 1 interreduces a finite set of nonzero polynomials.

Algorithm 1 Interreduction

Require: A finite set $Q \subseteq R \setminus \{0\}$ and a monomial order $\prec$
Ensure: An interreduced set that generates $\langle Q \rangle$

while there exist distinct $p, q \in Q$ and a term cu of p with $\mathrm{LM}(q) \mid u$ **do**
  $r \leftarrow p - \dfrac{cu}{\mathrm{LT}(q)} q$
  $Q \leftarrow Q \setminus \{p\}$
  if $r \neq 0$ **then**
    $Q \leftarrow Q \cup \{r\}$
  end if
end while
$Q \leftarrow \{p / \mathrm{LC}(p) : p \in Q\}$
return Q

The loop terminates. Let $\mu(Q)$ be the multiset of the monomials of all elements of Q. An iteration replaces the monomials of p by those of r, which lack u and are monomials of p or strictly smaller than u. If $r = 0$ or r already lies in Q, the iteration only removes the monomials of p. Either way, $\mu(Q)$ decreases in the multiset extension of $\prec$, the order generated by replacing an element of a multiset with finitely many smaller ones. It is well founded because $\prec$ is (Dershowitz and Manna, 1979). Interreduction preserves $\langle Q \rangle$ and $V(Q)$ (Cox et al., 2015, Ch. 2, §7), and it does not shrink the ideal generated by the leading monomials: an iteration with $u \neq \mathrm{LM}(p)$ keeps $\mathrm{LM}(p)$ as the leading monomial of r, and if $u = \mathrm{LM}(p)$, then $\mathrm{LM}(q)$ divides it and q stays in Q.

For a fixed monomial order, every ideal J has a unique interreduced Gröbner basis, its reduced Gröbner basis. The reduced basis is $\{1\}$ if and only if $J = \langle 1 \rangle$. Therefore, a Gröbner basis computation can describe a branch and detect when it is UNSAT (Cox et al., 2015; Kreuzer and Robbiano, 2000).

Reduction also tests containment. A polynomial g reduces to zero modulo a finite set G if repeatedly replacing g by $g - \frac{cu}{\mathrm{LT}(q)} q$, for a term cu of g and $q \in G$ with $\mathrm{LM}(q) \mid u$, reaches 0. If every element of a finite set G' reduces to zero modulo G, then $\langle G' \rangle \subseteq \langle G \rangle$, and hence $V(G) \subseteq V(G')$. If G is a Gröbner basis, the test is exact: $\langle G' \rangle \subseteq \langle G \rangle$ if and only if every element of G' reduces to zero modulo G (Cox et al., 2015, Ch. 2, §§3, 6).

2.4. Factor sets and hitting sets

Fix a factorization over K of each polynomial q of a finite system $Q \subseteq R \setminus \{0\}$:

$$q = c_q \prod_{f \in C_q} f^{e_{q,f}},$$

where $c_q \in K \setminus \{0\}$, the factor set C_q contains the distinct monic, nonconstant irreducible factors of q, and $e_{q,f} \geq 1$. We write $C(Q) = \bigcup_{q \in Q} C_q$. A point satisfies $q = 0$ exactly when at least one factor in C_q vanishes at that point.

A hitting set of Q is a set $H \subseteq C(Q)$ such that $H \cap C_q \neq \emptyset$ for every $q \in Q$. It is minimal if no proper subset is also a hitting set. We write $\mathcal{H}(Q)$ for the family of minimal hitting sets of Q. Each $H \in \mathcal{H}(Q)$ defines a branch in which all factors in H vanish (Berge, 1989).

3. Factor branching framework

Polynomial-ideal computations can be intractable even for small systems (Mayr and Meyer, 1982). Thus in this section we describe a method to break down a polynomial system into simpler branches which are easier for solvers to get a description of $\sqrt{\langle P \rangle}$ and $V(P)$.

3.1. Framework description

Algorithm 2 breaks the system P into smaller branches by factoring each generator and enumerating the minimal hitting sets of the factor sets. Each hitting set generates a new branch, which we interreduce and factor again, until it reaches a fixed point. We call a branch *terminal* if one round leaves it unchanged.

Hitting sets are minimal only among the children of one branch, so the loop of Algorithm 2 can end with distinct branches H and H' such that $V(H) \subseteq V(H')$. Solving H then adds no solutions. PRUNE(W) returns the branches whose solution sets it cannot certify to be contained in each other.

Algorithm 2 Refinement

Require: A finite polynomial system P
Ensure: The terminal branches of P
$W \leftarrow \mathcal{H}(P)$
repeat
  $W_{\text{prev}} \leftarrow W, \quad W \leftarrow \emptyset$
  for all $Q \in W_{\text{prev}}$ **do**
    $Q \leftarrow \textsc{Interreduce}(Q)$
    $W \leftarrow W \cup \mathcal{H}(Q)$
  end for
until $W = W_{\text{prev}}$
$W \leftarrow \textsc{Prune}(W)$ ▷ Optional
return W

Interreducing P first can expose factors that its generators hide, but interreducing can also eliminate factorizations without exposing new ones. So the decision on whether to interreduce first depends on P.

Let $\mathcal{L}(P)$ denote the terminal branches returned by Algorithm 2 on P. For each $H \in \mathcal{L}(P)$, the branch solver must either describe $V(H)$ exactly or certify that $V(H) = \emptyset$. We assume that every solver in this paper terminates and returns a correct output.

Algorithm 3 Factor branching

Require: A set of polynomials $P = \{p_1, \dots, p_m\}$ and a branch solver $\mathcal{B}$.
$S \leftarrow \emptyset$ ▷ Solution-set representations
for all $H \in \textsc{Refine}(P)$ **do**
  $S \leftarrow S \cup \mathcal{B}(H)$
end for
return S

A found branch solution can prune later branches. If this optimization is used, there are two reasons to start with branches that have fewer generators. First, they are likely to be solved faster. Secondly, they are likely to be higher dimensional and more likely to contain the probably lower dimensional later branches.

3.2. Theoretical analysis

Theorem 1 (One-round branch decomposition). *For every finite $Q \subseteq R \setminus \{0\}$,*

$$V(Q) = \bigcup_{H \in \mathcal{H}(Q)} V(H).$$

PROOF. Let $a \in V(Q)$. Every $q \in Q$ vanishes at a, so some factor in C_q does. The factors in $C(Q)$ that vanish at a therefore form a hitting set, which contains a minimal one H, and $a \in V(H)$. Conversely, let $a \in V(H)$ for some $H \in \mathcal{H}(Q)$. Every $q \in Q$ has a factor in H, which vanishes at a, so $a \in V(Q)$.

Theorem 2 (Termination of refinement). *Algorithm 2 terminates on every finite polynomial system.*

PROOF. One round replaces each branch Q by its children $\mathcal{H}(\widetilde{Q})$, where $\widetilde{Q} = \textsc{Interreduce}(Q)$, independently of the other branches. For a finite set G, write $M(G) = \langle \mathrm{LM}(g) : g \in G \rangle$.

We first show that every child H of Q is terminal or satisfies $M(\widetilde{Q}) \subsetneq M(\widetilde{H})$. Every $q \in \widetilde{Q}$ is a multiple of some $h \in H$, so $\mathrm{LM}(h) \mid \mathrm{LM}(q)$ and $M(\widetilde{Q}) \subseteq M(H)$. Suppose that equality holds, and let $h \in H$ divide $q \in \widetilde{Q}$. Since $\mathrm{LM}(h) \in M(\widetilde{Q})$, some $q' \in \widetilde{Q}$ has $\mathrm{LM}(q') \mid \mathrm{LM}(h) \mid \mathrm{LM}(q)$. Since $\widetilde{Q}$ is interreduced, $q' = q$, so q/h has leading monomial 1, and $h = q$ because both are monic. Every element of H is a factor of an element of $\widetilde{Q}$, and every element of $\widetilde{Q}$ has a factor in H, so $H = \widetilde{Q}$. Then H is interreduced and consists of monic irreducible polynomials, so $\widetilde{H} = H$ and $\mathcal{H}(H) = \{H\}$, that is, H is terminal. Otherwise $M(\widetilde{Q}) \subsetneq M(H) \subseteq M(\widetilde{H})$, since interreduction does not shrink the leading-monomial ideal.

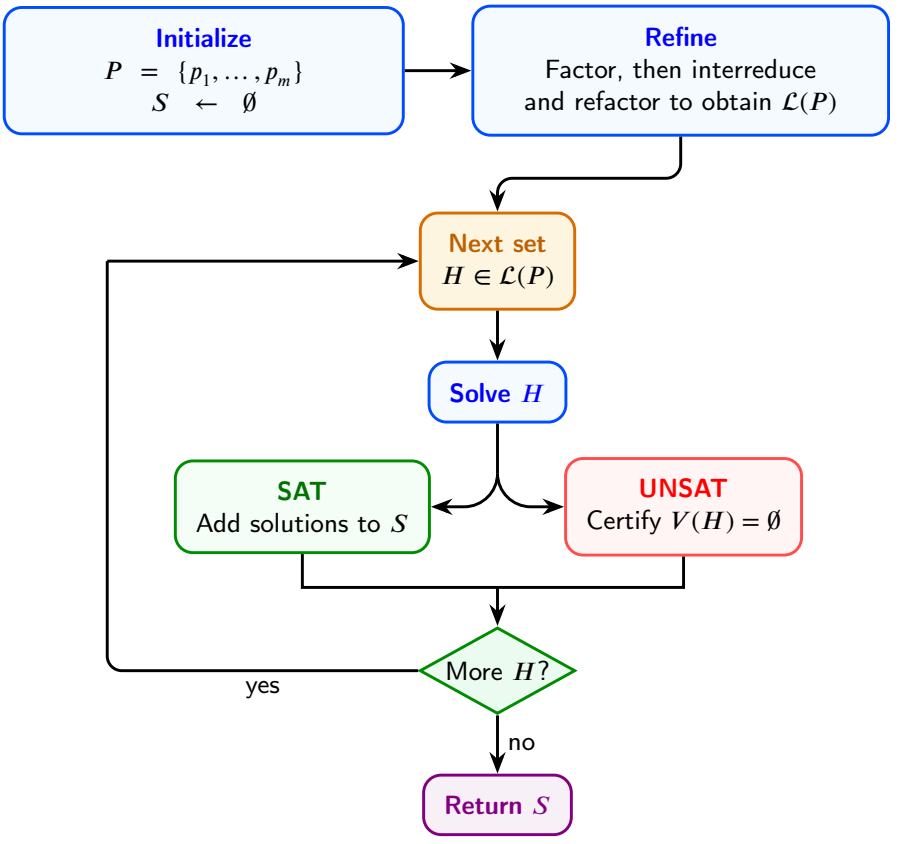


Figure 2: Factor branching procedure of Algorithm 3. Refinement produces terminal branches, which the exact solver either represents or certifies as UNSAT.

Next, we show that the rounds started from $\{Q\}$ reach a fixed point. Since R is Noetherian, it has no infinite strictly ascending chain of ideals, so we may assume this for every branch H with $M(\widetilde{Q}) \subsetneq M(\widetilde{H})$. A terminal Q is a fixed point. Otherwise, each of the finitely many children H of Q is terminal or has $M(\widetilde{Q}) \subsetneq M(\widetilde{H})$, so it reaches a fixed point after some k_H rounds, and $\{Q\}$ reaches one after $1 + \max_H k_H$ rounds. Since rounds act on each branch separately, $W = \mathcal{H}(P)$ reaches a fixed point, and the loop stops.

Finally, every branch of the final W is terminal. Otherwise, choose a nonterminal $Q \in W$ whose $M(\widetilde{Q})$ is minimal among the nonterminal branches of W. Since W is a fixed point, Q is a child of some $Q' \in W$. A terminal Q' has only the child Q', so Q' is nonterminal, and $M(\widetilde{Q'}) \subsetneq M(\widetilde{Q})$ contradicts the choice of Q.

Each child keeps at most one irreducible factor of each generator of its parent, which makes it a smaller and simpler system in the following sense.

Theorem 3 (Refined branches are simpler). *Let $H \in \mathcal{H}(\widetilde{Q})$ with $H \neq \widetilde{Q}$, where $\widetilde{Q} = \textsc{Interreduce}(Q)$. Each element of H divides a different element of $\widetilde{Q}$. The leading monomials of H are smaller than those of $\widetilde{Q}$ in the multiset extension of $\prec$, yet generate a strictly larger monomial ideal.*

Proof. Let $h \in H$. Since H is a minimal hitting set, $H \setminus \{h\}$ is not one, so some $q \in \widetilde{Q}$ has $C_q \cap H = \{h\}$. Let $\iota(h)$ be such a q. Then $h \mid \iota(h)$, and ι is injective, since $C_{\iota(h)} \cap H = \{h\}$ determines h. If $h \neq \iota(h)$, then $\iota(h)/h$ is not constant, since both polynomials are monic, so $\mathrm{LM}(h) \prec \mathrm{LM}(h)\,\mathrm{LM}(\iota(h)/h) = \mathrm{LM}(\iota(h))$, since $1 \prec v$ for every monomial $v \neq 1$ (Cox et al., 2015, Ch. 2, §4). The multiset of the leading monomials of H arises from that of $\widetilde{Q}$ by replacing $\mathrm{LM}(\iota(h))$ with $\mathrm{LM}(h)$ for every $h \in H$ and deleting $\mathrm{LM}(q)$ for every $q \in \widetilde{Q} \setminus \iota(H)$. Since $H \neq \widetilde{Q}$, some h has $h \neq \iota(h)$, or some q lies outside $\iota(H)$, so the multiset decreases strictly. Finally, with M as in the proof of Theorem 2, that proof shows that $M(\widetilde{Q}) \subseteq M(H)$, with equality only if $H = \widetilde{Q}$.

Lemma 1 (Refinement preserves solutions). *Every round of Algorithm 2 preserves* $\bigcup_{Q\in W} V(Q)$. *Consequently,*

$$V(P) = \bigcup_{H\in\mathcal{L}(P)} V(H) \qquad \text{and} \qquad \sqrt{\langle P\rangle} = \bigcap_{H\in\mathcal{L}(P)} \sqrt{\langle H\rangle}.$$

PROOF. Interreduction keeps $V(Q)$, so Theorem 1 gives $V(Q) = \bigcup_{H\in\mathcal{H}(\widetilde{Q})} V(H)$ for every branch Q, with $\widetilde{Q} =$ INTERREDUCE(Q), and a round keeps the union. The optional PRUNE removes only branches whose solution sets lie in that of a kept branch. The loop starts from $W = \mathcal{H}(P)$, so Theorem 1 and induction on the rounds give the first identity. By the Nullstellensatz, $\sqrt{\langle P\rangle} = I(V(P))$ and $\sqrt{\langle H\rangle} = I(V(H))$, and the vanishing ideal of a finite union is the intersection of the vanishing ideals, which gives the second.

Theorem 4 (Factor branching). *Algorithm 3 terminates and returns a decomposition of* $V(P)$*:*

$$V(P) = \bigcup_{H\in S} V(H) \qquad \text{and} \qquad \sqrt{\langle P\rangle} = \bigcap_{H\in S} \sqrt{\langle H\rangle}.$$

PROOF. By Theorem 2, REFINE(P) returns the finite set $\mathcal{L}(P)$, so the algorithm makes finitely many solver calls, each of which returns. The branches that $\mathcal{B}$ certifies UNSAT have empty solution sets, and $\mathcal{B}$ represents every other branch exactly, so the first identity follows from Lemma 1, and the second follows from the first by the Nullstellensatz, as in the proof of Lemma 1.

3.3. Minimal primes

Let $\mathcal{B}_{\min}$ be a branch solver which returns the minimal primes of its input. If we run Algorithm 3 with $\mathcal{B}_{\min}$ as a branch solver and apply the following postprocessing, we get the minimal primes of $\langle P\rangle$.

The output S is then the set of primes that $\mathcal{B}_{\min}$ returns on the branches in $\mathcal{L}(P)$, and the postprocessing keeps its minimal elements under inclusion in two steps.

1. Represent each prime by its reduced Gröbner basis. Equal primes have the same reduced Gröbner basis, so this merges the copies of a prime that several branches return.
2. Remove every $\mathfrak{p} \in S$ that contains another element $\mathfrak{p}'$ of S. The inclusion $\mathfrak{p}' \subseteq \mathfrak{p}$ holds exactly when every generator of $\mathfrak{p}'$ reduces to zero modulo the Gröbner basis of $\mathfrak{p}$.

A prime that strictly contains $\mathfrak{p}'$ has smaller dimension than $\mathfrak{p}'$, so the second step tests $\mathfrak{p}' \subseteq \mathfrak{p}$ only when $\mathfrak{p}'$ has the larger dimension.

Theorem 5 (Prime factor branching). *With the branch solver* $\mathcal{B}_{\min}$*, the output* S *of Algorithm 3 after the postprocessing is the minimal primes of* $\langle P\rangle$.

PROOF. By Theorem 2, $\mathcal{L}(P)$ is finite, so S is finite. On each $H \in \mathcal{L}(P)$, $\mathcal{B}_{\min}$ returns the minimal primes of $\langle H\rangle$, which intersect to $\sqrt{\langle H\rangle}$, so Lemma 1 gives

$$\bigcap_{\mathfrak{p}\in S} \mathfrak{p} = \bigcap_{H\in\mathcal{L}(P)} \sqrt{\langle H\rangle} = \sqrt{\langle P\rangle}.$$

In particular, every $\mathfrak{p} \in S$ contains $\langle P\rangle$.

Let $\mathfrak{q}$ be a minimal prime of $\langle P\rangle$. Every $\mathfrak{p} \in S$ with $\mathfrak{p} \subseteq \mathfrak{q}$ is a prime containing $\langle P\rangle$, so $\mathfrak{p} = \mathfrak{q}$ by the minimality of $\mathfrak{q}$. Such a $\mathfrak{p}$ exists, because a prime that contains a finite intersection of ideals contains one of them (Atiyah and Macdonald, 1969, Prop. 1.11), and $\mathfrak{q}$ contains $\bigcap_{\mathfrak{p}\in S} \mathfrak{p}$. Hence $\mathfrak{q} \in S$, no other element of S lies in $\mathfrak{q}$, and $\mathfrak{q}$ is a minimal element of S.

Conversely, let $\mathfrak{p}$ be a minimal element of S. It contains $\sqrt{\langle P\rangle}$, the intersection of the finitely many minimal primes of $\langle P\rangle$, so by the same proposition it contains one of them, $\mathfrak{q}$. By the previous paragraph $\mathfrak{q} \in S$, and the minimality of $\mathfrak{p}$ in S gives $\mathfrak{p} = \mathfrak{q}$.

4. Ideal-preserving factor branching

Algorithm 3 discards the exponents $e_{q,f}$ and hence only describes $V(P)$ and $\sqrt{\langle P\rangle}$. But ideal membership, multiplicities, and embedded components depend on $I = \langle P\rangle$ itself.

In this section, we keep I. We show that the SAT status of the branches, which Algorithm 3 already computes, certifies members of I. Replacing generators by these members gives a smaller system for the same ideal (Algorithm 4). We then decompose I into primary components,

$$I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_t.$$

The minimal primes of Theorem 5 give the isolated components, so a general primary decomposition runs only when I has embedded components, and they sort its components into isolated and embedded ones (Algorithm 5).

4.1. Ideal reduction

For a finite system Q and $h \in C(Q)$, let $a_h = \max\{e_{q,h} : q \in Q,\ h \in C_q\}$ be the largest exponent of h.

Suppose that a factor h lies in every SAT minimal hitting set of Q. Then h vanishes on all of $V(Q)$, and the Nullstellensatz puts some power of h in $\langle Q\rangle$. Lemma 2 shows that the power h^{a_h} suffices, so the factorization bounds the exponent without a Gröbner basis of $\langle Q\rangle$.

For example, $Q = \{xy,\ xz,\ yz - 1\}$ is interreduced for every monomial order. Its factor sets $\{x, y\}$, $\{x, z\}$, and $\{yz-1\}$ have the minimal hitting sets $\{x, yz-1\}$ and $\{y, z, yz-1\}$. The second is UNSAT, since $y = z = 0$ contradicts $yz = 1$. Hence x lies in every SAT minimal hitting set, and $a_x = 1$. Indeed, the cofactors y and z of x have no common zero on $V(Q)$, and multiplying $1 = z \cdot y - (yz - 1)$ by x gives

$$x = z \cdot xy - x\,(yz - 1) \in \langle Q\rangle.$$

Algorithm 4 adds these powers to the generators, drops the generators they divide, and repeats on the new system (Figure 3). We say that $\mathcal{B}$ certifies $V(H) = \emptyset$ when Algorithm 3 with $\mathcal{B}$ certifies every terminal branch of H as UNSAT. A minimal hitting set whose status is unknown, for example after a time limit, counts as SAT, which can only shrink Δ. If no minimal hitting set is SAT, the intersection is $C(Q)$, and Lemma 2 still applies. On the example, the first round replaces xy and xz by x, and the second round finds no new member and returns $\{x, yz - 1\}$.

Algorithm 4 Ideal reduction

Require: A finite polynomial system P and a branch solver $\mathcal{B}$
Ensure: A finite generating set Q of $\langle P\rangle$

```
Q ← INTERREDUCE(P)
loop
    H_sat ← {H ∈ H(Q) : B does not certify V(H) = ∅}
    Δ ← {h^{a_h} : h ∈ ⋂ H_sat} \ Q
    if Δ = ∅ then
        return Q
    end if
    Q ← INTERREDUCE(Δ ∪ {q ∈ Q : no element of Δ divides q})
end loop
```

The output generates $\langle P\rangle$, with the powers that the rounds find in place of the products they divide. When a single minimal hitting set H is SAT and every exponent is 1, one round gives $\langle P\rangle = \langle H\rangle$, as in the example. The ideal-level and the variety-level descriptions then coincide, and a Gröbner basis of $\langle P\rangle$ costs no more than one of the branch.

Each round needs only the intersection of the SAT minimal hitting sets, so an implementation need not classify every minimal hitting set. It keeps the factors F that lie in every SAT hitting set checked so far, starting from $F = C(Q)$, skips a hitting set that contains all of F, and stops checking once F is empty. Each checked SAT hitting set replaces F by its intersection with F, and each skipped one contains F, so the final F is $\bigcap \mathcal{H}_{\text{sat}}$. Two disjoint SAT hitting sets already empty F, so a round that finds nothing often costs only a few solver calls.

A polynomial u that must not vanish, such as κ_h in Subsection 5.4, changes only the SAT test: a minimal hitting set H counts as SAT when $V(H) \not\subseteq V(u)$. The proof of Lemma 2 then shows that Q and the cofactors of h have no

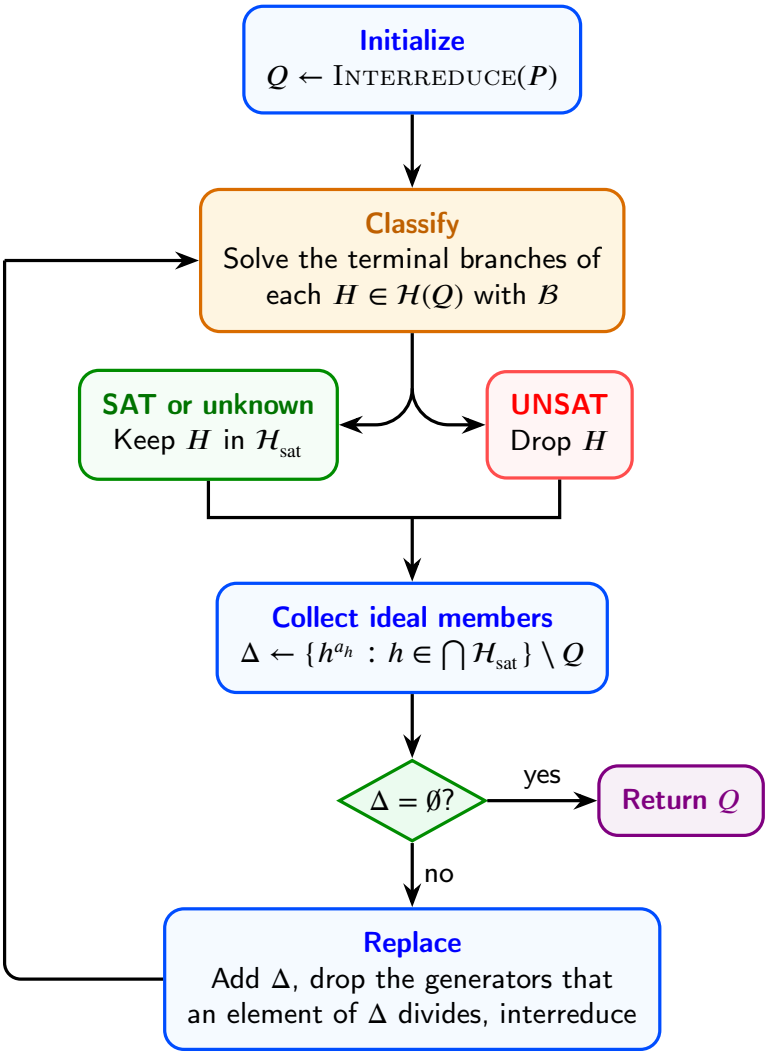


Figure 3: Ideal reduction of Algorithm 4. Every factor h chosen by all SAT minimal hitting sets enters the system as h^{a_h}, which replaces the generators it divides. The loop stops when no new power appears, and $\langle Q \rangle = \langle P \rangle$ throughout.

common zero outside $V(u)$, so the Nullstellensatz gives $u^s h^{a_h} \in \langle Q \rangle$ for some $s \geq 0$. Each round then keeps $\langle Q \rangle : u^\infty$ instead of $\langle Q \rangle$, and the termination argument of Theorem 6 is unchanged. Algorithm 5 then decomposes $\langle Q \rangle : u^\infty$, whose minimal primes are those of $\langle Q \rangle$ that do not contain u.

4.2. Primary factor branching

Algorithm 3 with $\mathcal{B}_{\min}$ describes only $\sqrt{I}$. A minimal primary decomposition of I (Subsection 2.2) also records the multiplicity of each component and the embedded components.

Algorithm 5 (Figure 4) computes the minimal primary decomposition. Let $\mathcal{B}_{\text{prim}}$ be an algorithm that returns a minimal primary decomposition of its input, for instance (Gianni et al., 1988; Shimoyama and Yokoyama, 1996).

The algorithm first replaces P by the output Q of Algorithm 4. By Theorem 6, $\langle Q \rangle = I$, so every later step works on I, and $\mathcal{B}_{\text{prim}}$ starts from a smaller input when Algorithm 4 replaces products by powers of their factors.

It then computes the minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ of I with Algorithm 3 on Q, the branch solver $\mathcal{B}_{\min}$, and the postprocessing of Subsection 3.3. The minimal hitting sets of Q that the last round of Algorithm 4 certified as UNSAT need not be revisited.

Optionally, the minimal primes give the isolated components without $\mathcal{B}_{\text{prim}}$. For each $\mathfrak{p}_i$, a *separator* is a polynomial $w_i \in \bigcap_{j \neq i} \mathfrak{p}_j \setminus \mathfrak{p}_i$. One exists because $\mathfrak{p}_i$ contains no other minimal prime and hence, by Subsection 2.2, not

their intersection. Saturating by w_i removes every component whose radical does not contain $\mathfrak{p}_i$ (Shimoyama and Yokoyama, 1996), and the top-dimensional part of $I : w_i^\infty$, the intersection of its primary components of maximal dimension (Eisenbud et al., 1992), is the isolated $\mathfrak{p}_i$-primary component $\mathfrak{q}_i$ (Lemma 3). Both computations start from a Gröbner basis G of I. If $\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r = I$, which G decides, then I has no embedded components, and the algorithm returns $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ without running $\mathcal{B}_{\text{prim}}$.

Otherwise, $\mathcal{B}_{\text{prim}}$ decomposes $\langle Q \rangle$, which is I.

A component $\mathfrak{q}$ is isolated if $\sqrt{\mathfrak{q}}$ is one of the $\mathfrak{p}_i$ and embedded otherwise. The radical of an embedded component strictly contains at least one $\mathfrak{p}_i$, so the variety of the component lies inside an irreducible component of $V(I)$.

For example, $I = \langle x^2, xy \rangle$ has the single minimal prime $\langle x \rangle$, which needs no separator. The top-dimensional part of I is $\langle x \rangle \neq I$, so $\mathcal{B}_{\text{prim}}$ runs and returns $I = \langle x \rangle \cap \langle x^2, y \rangle$. The component $\langle x \rangle$ is isolated, and $\langle x^2, y \rangle$ is embedded, since its radical $\langle x, y \rangle$ strictly contains $\langle x \rangle$.

Algorithm 5 Primary factor branching

Require: A finite polynomial system P, branch solvers $\mathcal{B}$ and $\mathcal{B}_{\text{min}}$, and a minimal primary decomposition algorithm $\mathcal{B}_{\text{prim}}$
Ensure: A minimal primary decomposition of $\langle P \rangle$, each component marked isolated or embedded

$Q \leftarrow$ IdealReduce$(P, \mathcal{B})$ ▷ Algorithm 4, $\langle Q \rangle = \langle P \rangle$
$\{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\} \leftarrow$ the minimal primes of $\langle Q \rangle$ ▷ Theorem 5
for all $i \in \{1, \ldots, r\}$ **do** ▷ Optional
  $w_i \leftarrow$ a separator of $\mathfrak{p}_i$
  $\mathfrak{q}_i \leftarrow$ the top-dimensional part of $\langle Q \rangle : w_i^\infty$
end for
if $\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r = \langle Q \rangle$ **then** ▷ Optional
  return $\{\mathfrak{q}_1, \ldots, \mathfrak{q}_r\}$, all marked isolated
end if
$\mathcal{D} \leftarrow \mathcal{B}_{\text{prim}}(Q)$
for all $\mathfrak{q} \in \mathcal{D}$ **do**
  mark $\mathfrak{q}$ isolated if $\sqrt{\mathfrak{q}} \in \{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\}$, and embedded otherwise
end for
return $\mathcal{D}$

A separator also needs no intersection of primes. The saturation $I : \mathfrak{p}_i^\infty = \{g \in R : \mathfrak{p}_i^m g \subseteq I \text{ for some } m \geq 0\}$ keeps exactly the components whose radicals do not contain $\mathfrak{p}_i$, so it lies in every other minimal prime but not in $\mathfrak{p}_i$. Hence every element of its Gröbner basis that does not reduce to zero modulo a Gröbner basis of $\mathfrak{p}_i$ is a separator. Which of the two constructions is cheaper depends on the ideal.

A zero-dimensional I needs no separators. Its components are pairwise comaximal, so $\dim_K R/I$ is the sum of the $\dim_K R/\mathfrak{q}_i$, each at least $\dim_K R/\mathfrak{p}_i$ with equality only if $\mathfrak{q}_i = \mathfrak{p}_i$. If $\dim_K R/I = \sum_i \dim_K R/\mathfrak{p}_i$, every component is therefore prime. Otherwise $\mathfrak{q}_i = I + \mathfrak{p}_i^m$ for the first m at which $\dim_K R/(I + \mathfrak{p}_i^m)$ stops growing, by Nakayama's lemma (Atiyah and Macdonald, 1969, Prop. 2.6). All primes of a zero-dimensional ideal are maximal, so it has no embedded components, and the intersection test can be skipped.

The mark needs no equality test of ideals. A prime that strictly contains $\mathfrak{p}_i$ has smaller dimension (Subsection 3.3), so $\mathfrak{q}$ is isolated exactly when $\sqrt{\mathfrak{q}}$ contains some $\mathfrak{p}_i$ of its own dimension. Containment means that the generators of $\mathfrak{p}_i$ reduce to zero modulo a Gröbner basis of $\sqrt{\mathfrak{q}}$.

Two properties of the output can also be checked independently. The components must intersect to I, which a Gröbner basis of $\langle P \rangle$ decides, and the isolated components must have exactly $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ as their radicals.

4.3. Theoretical analysis

Lemma 2 (SAT branches). *Let $\mathcal{H}_{\text{sat}} \subseteq \mathcal{H}(Q)$ contain every SAT minimal hitting set, and let $h \in C(Q)$ lie in every $H \in \mathcal{H}_{\text{sat}}$. Then $h^{a_h} \in \langle Q \rangle$.*

Proof. For $q \in Q$ with $h \in C_q$, let $r_q = q/h^{e_{q,h}}$ be the cofactor of h. We first show that the cofactors have no common zero on $V(Q)$. Suppose that $b \in V(Q)$ is a common zero of the cofactors. Then every $q \in Q$ has a factor $f \neq h$ with $f(b) = 0$: a factor of r_q if $h \in C_q$, and any factor that vanishes at b otherwise. So the factors $f \neq h$ in $C(Q)$ that

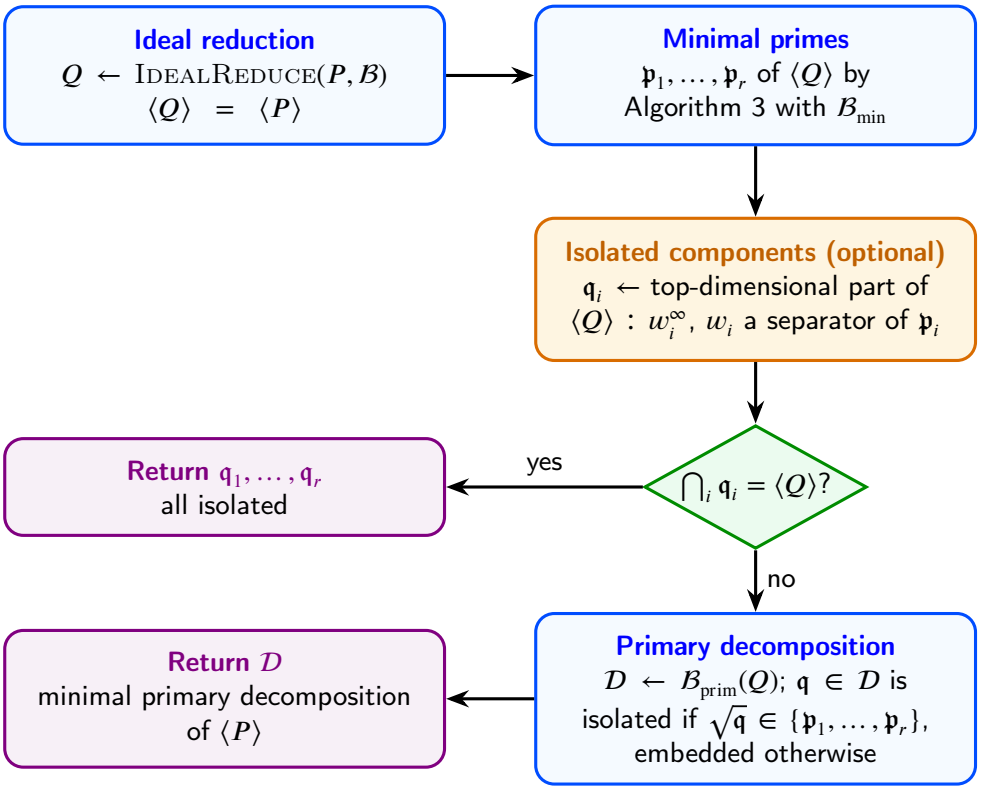


Figure 4: Primary factor branching of Algorithm 5. Ideal reduction shrinks the system to Q without changing the ideal. The separators of the minimal primes give the isolated components, which are returned if they intersect to $\langle Q \rangle$. Otherwise the general solver $\mathcal{B}_{\text{prim}}$ decomposes Q, and the minimal primes mark each component isolated or embedded.

vanish at b form a hitting set, and a minimal hitting set H inside it has $b \in V(H)$ and $h \notin H$. Then H is SAT, hence $H \in \mathcal{H}_{\text{sat}}$ and $h \in H$, a contradiction. By the Nullstellensatz, $1 = \sum_{q \in Q} g_q\, q + \sum_{q:\, h \in C_q} g'_q\, r_q$ for some $g_q, g'_q \in R$. Since $e_{q,h} \le a_h$, every $h^{a_h} r_q = h^{a_h - e_{q,h}} q$ lies in $\langle Q \rangle$, and multiplying by h^{a_h} gives $h^{a_h} \in \langle Q \rangle$.

Theorem 6 (Ideal reduction). *Algorithm 4 terminates and returns a finite generating set Q of $\langle P \rangle$.*

Proof. Every round keeps $\langle Q \rangle$. Indeed, $\mathcal{B}$ certifies $V(H) = \emptyset$ only when it holds (Lemma 1), so $\mathcal{H}_{\text{sat}}$ contains every SAT minimal hitting set. Hence $\Delta \subseteq \langle Q \rangle$ by Lemma 2, the dropped generators are multiples of elements of Δ, and interreduction keeps the ideal.

For termination, let $M(G) = \langle \mathrm{LM}(g) : g \in G \rangle$ as in the proof of Theorem 2. Consider a round with $\Delta \neq \emptyset$, and let $Q_0 = \Delta \cup \{q \in Q : \text{no element of } \Delta \text{ divides } q\}$, so that the round ends with $\textsc{Interreduce}(Q_0)$. At the start of every round, Q is interreduced, and in particular monic. Every $q \in Q$ is kept or is a multiple of an element of Δ, so $M(Q) \subseteq M(Q_0)$. Let $\delta = h^{a_h} \in \Delta$ and let $q \in Q$ have $e_{q,h} = a_h$, so that $\delta \mid q$. If $\mathrm{LM}(\delta) \in M(Q)$, some $q' \in Q$ has $\mathrm{LM}(q') \mid \mathrm{LM}(\delta) \mid \mathrm{LM}(q)$, so $q' = q$ since Q is interreduced. Then q/δ has leading monomial 1, and $\delta = q \in Q$ since both are monic, which contradicts $\delta \notin Q$. Hence $M(Q) \subsetneq M(Q_0)$, and since interreduction does not shrink the leading-monomial ideal, $M(Q)$ grows strictly in every round that does not return. Since R is Noetherian, there are finitely many rounds, and each makes finitely many calls of $\mathcal{B}$.

Lemma 3 (Isolated components by separators). *Let $\mathfrak{p}$ be a minimal prime of an ideal I and w a separator of $\mathfrak{p}$. Then $\mathfrak{p}$ is the only minimal prime of $I : w^{\infty}$, and the $\mathfrak{p}$-primary component of I is the intersection of the primary components of $I : w^{\infty}$ of maximal dimension.*

Proof. Let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_t$ be a minimal primary decomposition with $\mathfrak{p}_j = \sqrt{\mathfrak{q}_j}$, and let $T = \{j : w \notin \mathfrak{p}_j\}$. By Subsection 2.2, $I : w^{\infty} = \bigcap_{j \in T} \mathfrak{q}_j$, and this decomposition is again minimal, since a $\mathfrak{q}_j$ that contained the intersection

of the other kept components would contain the smaller intersection of all other components. For $j \in T$, the prime $\mathfrak{p}_j$ contains I, hence $\sqrt{I}$, the intersection of the minimal primes of I, and hence one of them (Atiyah and Macdonald, 1969, Prop. 1.11). This minimal prime is $\mathfrak{p}$, since every other one contains w. The $\mathfrak{p}$-primary component $\mathfrak{q}$ of I is kept, since $w \notin \mathfrak{p}$, so $\mathfrak{p}$ contains $I : w^\infty$. Every prime that contains $I : w^\infty$ contains some kept $\mathfrak{q}_j$ (Atiyah and Macdonald, 1969, Prop. 1.11), hence $\mathfrak{p}_j \supseteq \mathfrak{p}$, so $\mathfrak{p}$ is the only minimal prime of $I : w^\infty$.

It follows that $\mathfrak{q}$ is the isolated $\mathfrak{p}$-primary component of $I : w^\infty$, which is the same in every minimal primary decomposition of $I : w^\infty$. Every other component of such a decomposition has a radical that contains $I : w^\infty$, hence $\mathfrak{p}$, and differs from $\mathfrak{p}$. A prime that strictly contains $\mathfrak{p}$ has smaller dimension (Subsection 3.3), so $\mathfrak{q}$ is the only component of maximal dimension.

Theorem 7 (Primary factor branching). *Algorithm 5, with or without its optional lines, terminates and returns a minimal primary decomposition $\mathcal{D}$ of $\langle P \rangle$:*

$$\langle P \rangle = \bigcap_{\mathfrak{q} \in \mathcal{D}} \mathfrak{q}.$$

It marks a component isolated exactly when the component is an isolated component of $\langle P \rangle$, and the radical of every embedded component strictly contains a minimal prime of $\langle P \rangle$. If every component is given by a Gröbner basis, then $g \in \langle P \rangle$ if and only if g reduces to zero modulo the basis of every component.

PROOF. By Theorem 6, Algorithm 4 terminates and returns Q with $\langle Q \rangle = \langle P \rangle = I$, so by Theorem 5 the $\mathfrak{p}_i$ are the minimal primes of I. The remaining lines make finitely many solver calls, so the algorithm terminates.

Suppose that the optional lines return. By Lemma 3, each $\mathfrak{q}_i$ is the isolated $\mathfrak{p}_i$-primary component of I, and $\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r = I$. The radicals $\mathfrak{p}_i$ are distinct, and no $\mathfrak{q}_i$ is redundant. Otherwise $\mathfrak{p}_i$ contains $\bigcap_{j \neq i} \mathfrak{q}_j$, hence some $\mathfrak{q}_j$ with $j \neq i$ (Atiyah and Macdonald, 1969, Prop. 1.11), and hence $\mathfrak{p}_j$, which contradicts the minimality of $\mathfrak{p}_i$. So the $\mathfrak{q}_i$ form a minimal primary decomposition of I, and all of them are isolated.

Otherwise, $\mathcal{B}_{\text{prim}}$ decomposes $\langle Q \rangle = I$. A component is isolated exactly when its radical is a minimal prime of I, that is, one of the $\mathfrak{p}_i$, which is the mark. The radical of an embedded component is a prime that contains I, hence some $\mathfrak{p}_i$ as in the proof of Lemma 3, and it differs from $\mathfrak{p}_i$.

In both cases, $g \in \langle P \rangle$ if and only if g lies in every component, and a Gröbner basis decides membership by reduction (Subsection 2.3).

5. Experiments

5.1. Setup

We implemented Algorithm 3 in Rust 1.98.1 and used Symbolica 3.0.1 for polynomial arithmetic over the rationals (Ruijl, 2026). Refinement applies the pruning of Subsection 3.1, and the branch solver processes the terminal branches fewest generators first. In all experiments, we solve the branches with a Gröbner-basis computation. The available implementations are M4GB, F5, GVW, F4, and Singular 4.4.1 (Decker et al., 2025; Faugère, 1999, 2002; Gao et al., 2016; Makarim and Stevens, 2017). A basis containing 1 certifies that the branch is UNSAT. The comparison also includes six direct whole-ideal Gröbner-basis configurations and three decomposition methods from Singular and Macaulay2 1.26.06 (Decker et al., 2025; Grayson and Stillman, 2026).

We ran the benchmark on an Apple M4 Pro with 14 CPU cores and 24 GB of memory. Every table entry is a single run. Every method received 60 seconds per system.

For factor branching, T is end-to-end wall time and F is the time to first SAT branch, measured from the start of the minimal-hitting-set search. Factorization and input processing are therefore included in T but not in F. The complete measurements are in the `results` directory of the repository listed under Data availability.

5.2. Baseline systems

The baseline has 20 generated systems in six families.

- Signed-distance cycles and affine-factor hypergraphs test many inexpensive branches.
- The overlapping random multiples use seeded dense quadratic factors that recur across equations, with no factor shared by every equation.

- For gated systems, we create a new variable and multiply every equation in Cyclic, Katsura, or dense quadratic by it (Björck and Fröberg, 1991; Böge et al., 1986).
- This produces one easy selector branch and one nonlinear residual branch.

Family	Tested sizes	Variables/equations; hitting sets
Signed-distance cycle	$n = 12, 14, 16$	$12/12, 14/14, 16/16$; $924, 3432, 12870$
Overlapping random multiples	$(n, p, k) = (4, 6, 2), (6, 8, 3), (8, 10, 3)$	n/n; $4, 16, 28$
Positive affine hypergraph	$(n, m, k) = (8, 6, 4), (10, 8, 5), (12, 10, 6)$	n/m; $63, 239, 833$
Gated Cyclic	orders $5, 6, 7$	$6/5, 7/6, 8/7$; 2 each
Gated Katsura and dense quadratics	Katsura $6, 7, 8$; dense $n = 5, 6, 7$	$8/7, 9/8, 10/9, 6/5, 7/6, 8/7$; 2 each
Square affine hypergraph	$(n, k) = (8, 4), (10, 5)$	n/n; $178, 309$

Table 1
The generated systems used for the baseline benchmark.

Table 2 selects the fastest completed factor branching backend and the fastest completed method in each baseline class. A factor branching entry gives T/F in seconds; the other entries give total wall time. "Timeout" means that every tested method in that class timed out.

System	Best factor branching T/F	Best direct GB	Best decomposition
Signed distance, $n = 12$	F5 2.669/0.006	**F4 0.443**	primdec 3.056
Signed distance, $n = 14$	GVW 9.824/0.006	**Singular 7.421**	primdec 29.894
Signed distance, $n = 16$	**M4GB 37.391/0.006**	timeout	timeout
Overlapping $(4, 6, 2)$	**F5 0.014/0.003**	F4 0.429	facstd 0.265
Overlapping $(6, 8, 3)$	**F5 0.160/0.006**	timeout	timeout
Overlapping $(8, 10, 3)$	**Singular 2.671/0.013**	timeout	timeout
Positive affine $(8, 6, 4)$	**M4GB 0.180/0.003**	timeout	facstd 0.368
Positive affine $(10, 8, 5)$	**F4 0.703/0.003**	timeout	facstd 1.836
Positive affine $(12, 10, 6)$	**F4 2.398/0.003**	timeout	facstd 53.318
Gated Cyclic-5	F4 0.011/0.003	**GVW 0.009**	facstd 0.181
Gated Cyclic-6	**Singular 0.043/0.010**	Singular 0.309	M2 2.620
Gated Cyclic-7	timeout	timeout	timeout
Gated Katsura-6	**Singular 0.047/0.010**	GVW 0.226	facstd 0.233
Gated Katsura-7	**Singular 0.283/0.010**	GVW 3.176	facstd 0.539
Gated Katsura-8	**Singular 2.575/0.010**	timeout	facstd 3.596
Gated dense, $n = 5$	**Singular 0.025/0.010**	M4GB 0.171	facstd 0.198
Gated dense, $n = 6$	**Singular 0.552/0.010**	M4GB 4.736	facstd 0.755
Gated dense, $n = 7$	**Singular 42.670/0.010**	timeout	facstd 48.820
Square affine $(8, 4)$	**M4GB 0.532/0.003**	timeout	facstd 0.601
Square affine $(10, 5)$	**F4 0.920/0.003**	timeout	facstd 2.141
#systems solved	19	9	16
#fastest (ties counted)	16	3	0

Table 2
Best completed baseline results across solver classes. All times are in seconds.

Factor branching is a good choice when factorization produces branch ideals that are substantially easier than the expanded input ideal. A large number of hitting sets can still be manageable when individual branches are cheap to solve; the positive affine family shows that the hitting-set count alone is not decisive. The method is less effective when one branch retains the difficulty of the original nonlinear system, as Cyclic-7 does.

The measurements of time to first SAT branch support this explanation. On positive affine $(12, 10, 6)$, F is 0.003 seconds despite the 833 hitting sets. On gated Katsura-8, the first SAT branch is found in 0.010 seconds, but the full run

takes 2.575 seconds. Factorization therefore finds the easy component quickly, while the nonlinear residual controls completion time.

5.3. Size-graded families

The baseline families were built to have factor structure. The following five families inherit it from the problem instead, and we grow each until every method times out.

- The reduced cyclic n-roots systems `Singular.rcyclic_n`, a variant of the cyclic n-roots problem (Björck and Fröberg, 1991), taken unchanged from SymbolicData[1].
- Nash equilibria of n-player games with two pure strategies per player (Sturmfels, 2002). Each player i either plays a pure strategy, $p_i(p_i - 1) = 0$, or is indifferent, so its payoff difference vanishes. The payoffs are random integers in $[-9, 9]$ (generic) or in $\{-1, 0, 1\}$ (degenerate).
- Critical points of a random quadratic program f on every face of the box $[0, 1]^n$, given by $x_i(x_i - 1)\,\partial_i f = 0$.
- Equilibria of a random Lotka–Volterra community on every species support (Hofbauer and Sigmund, 1998), given by $x_i\big(r_i + \sum_j a_{ij}x_j\big) = 0$ with $a_{ii} < 0$.

Each size is a single seeded instance. The setup is that of Subsection 5.1 with four differences. Factor branching entries give T only. The direct class adds Singular's `std` in degrevlex (Singular dp) and omits standalone F4, which remains a factor branching backend. A method that times out at one size is not run at larger sizes.

Table 3 shows the same pattern in every family. The baselines win only on small systems, each of which they finish in under 0.5 seconds. From rcyclic-10, $n = 4$ for the Nash games, $n = 7$ for box-QP, and $n = 10$ for Lotka–Volterra, factor branching is fastest, and its lead grows with n: on rcyclic it rises from 2.3 times at $n = 10$ to 372 times at $n = 18$. Only factor branching completes rcyclic-19 and box-QP with $n = 8$. The two families reach this point differently. Every rcyclic system refines into only 3 terminal branches, each far easier than the whole system. Box-QP refines into up to 3^n branches, 6534 at $n = 8$, and each branch is cheap, as in the positive affine family.

Factor branching can also use a decomposition method as its branch solver. Table 4 compares `facstd` as a backend with `facstd` alone. It also compares each new algorithm with the Singular command that computes the same object: Algorithm 5 with primdec, and Algorithm 3 with `minAssGTZ`. Both members of these two pairs run in one Python process that drives one Singular process for the whole run. Refinement interreduces in lex order, and every Gröbner basis is in degrevlex. The implementations use the optimizations described after each algorithm. Algorithm 3 with `minAssGTZ` interreduces P first, solves the branches fewest generators first, lets only positive-dimensional primes prune later branches, and prunes a branch only if it contains another as a set. Its branch solver $\mathcal{B}_{\min}$ computes a factorizing Gröbner basis (Gräbe, 1995), returns each linear piece as a prime, and decomposes every other piece with `minAssGTZ` (Gianni et al., 1988). Algorithm 5 runs Algorithm 4 with the factorizing Gröbner basis as its branch solver, takes the minimal primes of the reduced system from Algorithm 3 with `minAssGTZ`, and uses the separators and the zero-dimensional shortcut of Subsection 4.2. It runs primdec on the reduced system only when the separators detect an embedded component, and marks its components with the minimal primes.

Factor branching with `facstd` completes every rcyclic system in under 0.3 seconds, whereas `facstd` alone times out from rcyclic-10. It also completes the largest Nash and Lotka–Volterra systems that factor branching solves, on which `facstd` alone times out. On box-QP, the $n = 4$ Nash games, and the smaller Lotka–Volterra systems, `facstd` alone is faster. There, factor branching starts Singular once per terminal branch, whereas `facstd` alone starts it once.

Algorithm 3 with `minAssGTZ` returns the minimal primes of every listed system up to rcyclic-14, the Nash games with $n = 4$, box-QP with $n = 9$, and Lotka–Volterra with $n = 14$, whereas `minAssGTZ` alone stops after rcyclic-8, box-QP with $n = 6$, Lotka–Volterra with $n = 6$, and the $n = 3$ Nash games. On the largest box-QP and Lotka–Volterra systems, it costs at most about 1.5 times as much as factor branching with `std` in the same harness, for example 23.05 against 15.13 seconds for box-QP with $n = 9$. It stops when `minAssGTZ` cannot decompose one terminal branch. On rcyclic-13, a single branch takes almost all of the 18 seconds, and on the degenerate Nash game with $n = 5$, `minAssGTZ` does not finish the branch on which every player is indifferent.

Algorithm 5 completes 8 of the listed systems and primdec alone 7, and where both finish they return the same components. It is faster on box-QP with $n = 5$ and $n = 6$, 0.715 against 16.20 seconds at $n = 5$, and only it

[1] `https://github.com/symbolicdata/data`, commit `a5cdf53`.

System	Best factor branching T	Best direct GB	Best decomposition
rcyclic-8	F4 0.019	**GVW 0.008**	facstd 0.425
rcyclic-10	**F4 0.018**	M4GB 0.041	timeout
rcyclic-12	**F4 0.021**	M4GB 0.198	timeout
rcyclic-14	**F4 0.029**	Singular dp 0.676	timeout
rcyclic-16	**Singular 0.062**	Singular dp 7.154	timeout
rcyclic-17	**Singular 0.091**	M2 19.76	timeout
rcyclic-18	**Singular 0.093**	M2 34.60	timeout
rcyclic-19	**Singular 0.113**	timeout	timeout
Nash generic, $n = 3$	F5 0.044	**F5 0.006**	facstd 0.197
Nash generic, $n = 4$	**GVW 0.136**	Singular dp 0.262	facstd 0.262
Nash generic, $n = 5$	**Singular 1.382**	timeout	timeout
Nash generic, $n = 6$	timeout	timeout	timeout
Nash degenerate, $n = 3$	M4GB 0.043	**GVW 0.005**	facstd 0.179
Nash degenerate, $n = 4$	**F5 0.086**	Singular lex 0.215	facstd 0.203
Nash degenerate, $n = 5$	**Singular 0.714**	timeout	timeout
Nash degenerate, $n = 6$	timeout	timeout	timeout
Box-QP, $n = 4$	F4 0.237	**F5 0.005**	facstd 0.179
Box-QP, $n = 5$	F5 0.700	**Singular dp 0.216**	facstd 0.297
Box-QP, $n = 6$	M4GB 2.174	**F5 0.033**	primdec 0.939
Box-QP, $n = 7$	**F5 6.494**	timeout	facstd 12.63
Box-QP, $n = 8$	**F5 20.42**	timeout	timeout
Box-QP, $n = 9$	timeout	timeout	timeout
Lotka–Volterra, $n = 6$	M4GB 0.191	**F5 0.006**	facstd 0.216
Lotka–Volterra, $n = 8$	F4 0.809	Singular dp 0.701	**facstd 0.464**
Lotka–Volterra, $n = 10$	**GVW 3.002**	timeout	facstd 6.432
Lotka–Volterra, $n = 12$	**F5 13.15**	timeout	timeout
Lotka–Volterra, $n = 14$	timeout	timeout	timeout
#systems solved	27	20	13
#fastest (ties counted)	18	8	1

Table 3
Best completed results across solver classes on the size-graded families, with a 60-second limit. For rcyclic, only selected sizes are shown; the summary rows count all 31 systems, rcyclic-8 to rcyclic-19 included. All times are in seconds.

completes Lotka–Volterra with $n = 8$. On these three systems the separators show that no embedded component exists, so primdec never runs. When embedded components exist, it runs primdec on the reduced system and pays for both, 6.117 against 5.004 seconds on the degenerate Nash game with $n = 4$. On rcyclic-8, it finds the three isolated components in 0.37 seconds, one of them not prime, and detects an embedded component, but primdec times out, as it does alone. Algorithm 4 stops after one round on every system it finishes, so the reduced system is the interreduced input. Algorithm 5 needs a Gröbner basis of the ideal, so it times out wherever `std` does, at box-QP with $n = 7$, Lotka–Volterra with $n = 10$, and the Nash games with $n = 5$. On the generic Nash game with $n = 4$, computing its isolated components does not finish, although `std` does.

5.4. Physical System

The system is an order-five approximation constraining graviton couplings in asymptotic-safety models to remain consistent with quantum field theory (QFT) (Rafie-Zinedine, 2018; Vacca and Zanusso, 2010). Solving this system helps determine admissible fixed-point structures in quantum gravity (Vacca and Zanusso, 2010). It has 28 equations in 21 variables, and its coefficients involve π. One coupling, κ_h, must be nonzero. Factor branching and the Singular and Macaulay2 baselines impose this by saturating by κ_h; the standalone M4GB, F5, GVW, and F4 runs, which cannot saturate, add the Rabinowitsch generator $1 - s\kappa_h$ (Rabinowitsch, 1930).

	facstd		primdec		minAssGTZ	
System	FB	alone	Primary	alone	Prime	alone
rcyclic-8	**0.038**	0.425	timeout	timeout	**0.081**	10.64
rcyclic-10	**0.040**	timeout	timeout	timeout	**0.110**	timeout
rcyclic-12	**0.043**	timeout	timeout	timeout	**1.472**	timeout
rcyclic-14	**0.053**	timeout	timeout	timeout	**21.22**	timeout
rcyclic-16	**0.079**	timeout	timeout	timeout	timeout	timeout
rcyclic-17	**0.119**	timeout	timeout	timeout	timeout	timeout
rcyclic-18	**0.162**	timeout	timeout	timeout	timeout	timeout
rcyclic-19	**0.285**	timeout	timeout	timeout	timeout	timeout
Nash generic, $n = 3$	**0.138**	0.197	0.065	**0.063**	**0.063**	0.073
Nash generic, $n = 4$	0.449	**0.262**	timeout	timeout	**0.121**	timeout
Nash generic, $n = 5$	**13.48**	timeout	timeout	timeout	timeout	timeout
Nash degenerate, $n = 3$	**0.136**	0.179	0.087	**0.074**	**0.060**	0.069
Nash degenerate, $n = 4$	0.263	**0.203**	6.117	**5.004**	**0.088**	timeout
Nash degenerate, $n = 5$	**1.634**	timeout	timeout	timeout	timeout	timeout
Box-QP, $n = 4$	0.661	**0.179**	**0.089**	0.128	**0.089**	0.170
Box-QP, $n = 5$	1.928	**0.297**	**0.715**	16.20	**0.166**	17.00
Box-QP, $n = 6$	7.528	**1.259**	**0.423**	3.172	**0.446**	1.564
Box-QP, $n = 7$	22.77	**12.63**	timeout	timeout	**1.564**	timeout
Box-QP, $n = 8$	timeout	timeout	timeout	timeout	**5.742**	timeout
Box-QP, $n = 9$	timeout	timeout	timeout	timeout	**23.05**	timeout
Lotka–Volterra, $n = 6$	0.615	**0.216**	**0.101**	0.128	**0.096**	0.161
Lotka–Volterra, $n = 8$	3.012	**0.464**	**1.873**	timeout	**0.254**	timeout
Lotka–Volterra, $n = 10$	10.05	**6.432**	timeout	timeout	**1.059**	timeout
Lotka–Volterra, $n = 12$	**41.47**	timeout	timeout	timeout	**5.286**	timeout
Lotka–Volterra, $n = 14$	timeout	timeout	timeout	timeout	**31.61**	timeout

Table 4
`facstd` as the factor branching (FB) backend and alone, and each new algorithm against the Singular command with the same output, with a 60-second limit; the faster of each pair is in bold. Primary is primary factor branching (Algorithm 5), and Prime is prime factor branching, Algorithm 3 with `minAssGTZ`. Both members of these two pairs run in one Python process that drives one Singular process and exclude the input conversion, so their times are not comparable with the `facstd` columns or with Table 3. A method that timed out at a smaller size of a family was not run again and is listed as a timeout. Sizes at which every method times out are omitted, and for rcyclic only the sizes of Table 3 are shown. All times are in seconds.

For this larger system, every run, including the standalone baselines, has a 120-second total limit. Within it, factor branching gives its saturation of the system five seconds and places no further limit on a branch. Refinement removes κ_h from every factor set and drops a branch once κ_h reduces to zero modulo its generators, since neither step loses a point with $\kappa_h \neq 0$. The branch solver receives each terminal branch with the generator $1 - s\kappa_h$.

All nine standalone baselines time out after 120 seconds on the physical system: M4GB, F5, GVW, and F4 in grevlex order, Singular and Macaulay2 lex Gröbner bases, Singular `facstd` and `primdecGTZ`, and Macaulay2 `decompose`. Factor branching completes with every backend in 0.78 to 0.86 seconds. The saturation by κ_h takes about 0.75 seconds of this time; afterwards every backend shows all 10 terminal branches SAT within 0.11 seconds.

Without a time limit and without the optional separators, Algorithm 5 returns a minimal primary decomposition of the physical system over $\mathbb{Q}(\pi)$. Algorithm 4, with κ_h kept nonzero as in Subsection 4.1, reduces the 28 equations to 18 generators in three rounds and 0.6 seconds. The saturation of these generators by κ_h has 56 generators. Algorithm 3 with `minAssGTZ` finds the six minimal primes in 0.2 seconds, and `primdecSY` (Shimoyama and Yokoyama, 1996)

Backend	Factor branching T/F (s)	Factor branching result
M4GB	0.796/0.770	complete: 10 SAT, 0 UNSAT
F5	0.788/0.763	complete: 10 SAT, 0 UNSAT
GVW	0.786/0.755	complete: 10 SAT, 0 UNSAT
F4	0.781/0.755	complete: 10 SAT, 0 UNSAT
Singular	0.844/0.762	complete: 10 SAT, 0 UNSAT
`facstd`	0.857/0.764	complete: 10 SAT, 0 UNSAT

Table 5
Physical-system results with a 120-second total limit per run. Factor branching refines the saturated system into 10 terminal branches, which its result column counts by outcome; its T and F include the saturation. The nine standalone baselines, including standalone `facstd`, all time out and are omitted.

decomposes the saturation in 213 seconds. On the unreduced system, `primdecSY` did not finish in 30 minutes when $\kappa_h \neq 0$ was encoded by a Rabinowitsch generator. Six of the ten components are isolated, in two families of dimensions 13, 12, and 8. In one family they are prime. In the other, each component arises from its radical by replacing one generator, a variable, by its square, which doubles the degree. Each family also has two embedded components, one of dimension 10 inside its 12-dimensional component and one of dimension 8 inside its 13-dimensional component. A Gröbner basis of the saturation confirms that the components intersect to $\langle P \rangle : \kappa_h^{\infty}$. Every step uses lex order with the variables ordered by a min-fill heuristic on the supports of the equations. With the variables in alphabetical order, Algorithm 4 stops at 27 generators, because the factorizing Gröbner basis of the largest hitting set does not finish within 120 seconds.

6. Conclusion

We introduced a solver-agnostic framework that decomposes polynomial systems into smaller branches by enumerating minimal hitting sets of their factor sets.

We proved that refinement terminates and preserves the full solution set.

On the baseline benchmarks, factor branching completed 19 of 20 systems, compared with 9 for direct Gröbner-basis methods and 16 for decomposition methods. On the QFT system, where one coupling must be nonzero, imposing this by saturation lets factor branching complete with all six backends in under 0.9 seconds, while all nine standalone baselines time out. Primary factor branching decomposes the same saturated system in under four minutes.

Future work will test this benefit on other systems.

CRediT authorship contribution statement

Ilay Menahem: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Visualization. **Michael Shalyt:** Writing – review & editing. **Ido Kaminer:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code and data associated with this work are publicly available at `https://github.com/IlayMenahem/soothing`.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI tools to assist with drafting and editing portions of the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

Atiyah, M.F., Macdonald, I.G., 1969. Introduction to Commutative Algebra. Addison-Wesley, Reading, MA.

Berge, C., 1989. Hypergraphs: Combinatorics of Finite Sets. North-Holland, Amsterdam.

Björck, G., Fröberg, R., 1991. A faster way to count the solutions of inhomogeneous systems of algebraic equations, with applications to cyclic *n*-roots. Journal of Symbolic Computation 12, 329–336. doi:10.1016/S0747-7171(08)80153-8.

Böge, W., Gebauer, R., Kredel, H., 1986. Some examples for solving systems of algebraic equations by calculating Gröbner bases. Journal of Symbolic Computation 2, 83–98. doi:10.1016/S0747-7171(86)80014-1.

Buchberger, B., 2006. Bruno Buchberger's PhD thesis 1965: An algorithm for finding the basis elements of the residue class ring of a zero dimensional polynomial ideal. Journal of Symbolic Computation 41, 475–511. doi:10.1016/j.jsc.2005.09.007.

Cox, D.A., Little, J., O'Shea, D., 2015. Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra. 4 ed., Springer, Cham. doi:10.1007/978-3-319-16721-3.

Decker, W., Greuel, G.M., Pfister, G., Schönemann, H., 2025. Singular 4-4-1: A computer algebra system for polynomial computations. https://www.singular.uni-kl.de.

Dershowitz, N., Manna, Z., 1979. Proving termination with multiset orderings. Communications of the ACM 22, 465–476. doi:10.1145/359138.359142.

Eisenbud, D., Huneke, C., Vasconcelos, W., 1992. Direct methods for primary decomposition. Inventiones Mathematicae 110, 207–235. doi:10.1007/BF01231331.

Faugère, J.C., 1999. A new efficient algorithm for computing Gröbner bases (F4). Journal of Pure and Applied Algebra 139, 61–88. doi:10.1016/S0022-4049(99)00005-5.

Faugère, J.C., 2002. A new efficient algorithm for computing Gröbner bases without reduction to zero (F5), in: Proceedings of the 2002 International Symposium on Symbolic and Algebraic Computation, ACM. pp. 75–83. doi:10.1145/780506.780516.

Gao, S., Volny, IV, F., Wang, M., 2016. A new framework for computing Gröbner bases. Mathematics of Computation 85, 449–465. doi:10.1090/MCOM/2969.

Gianni, P., Trager, B., Zacharias, G., 1988. Gröbner bases and primary decomposition of polynomial ideals. Journal of Symbolic Computation 6, 149–167. doi:10.1016/S0747-7171(88)80040-3.

Gräbe, H.G., 1995. On factorized Gröbner bases, in: Fleischer, J., Grabmeier, J., Hehl, F.W., Küchlin, W. (Eds.), Computer Algebra in Science and Engineering. World Scientific, Singapore, pp. 77–89. URL: https://www.informatik.uni-leipzig.de/~graebe/ComputerAlgebra/Publications/fgb.pdf.

Grayson, D.R., Stillman, M.E., 2026. Macaulay2, a software system for research in algebraic geometry. https://macaulay2.com. Version 1.26.06.

Hofbauer, J., Sigmund, K., 1998. Evolutionary Games and Population Dynamics. Cambridge University Press, Cambridge.

Kera, H., Ishihara, Y., Kambe, Y., Vaccon, T., Yokoyama, K., 2024. Learning to compute Gröbner bases, in: Advances in Neural Information Processing Systems. URL: https://openreview.net/forum?id=ZRz7XlxBzQ.

Kreuzer, M., Robbiano, L., 2000. Computational Commutative Algebra 1. Springer, Berlin. doi:10.1007/978-3-540-70628-1.

Makarim, R.H., Stevens, M., 2017. M4GB: An efficient Gröbner-basis algorithm, in: Proceedings of the 2017 ACM International Symposium on Symbolic and Algebraic Computation, ACM. pp. 293–300. doi:10.1145/3087604.3087638.

Mayr, E.W., Meyer, A.R., 1982. The complexity of the word problems for commutative semigroups and polynomial ideals. Advances in Mathematics 46, 305–329. doi:10.1016/0001-8708(82)90048-2.

Peifer, D., Stillman, M., Halpern-Leistner, D., 2020. Learning selection strategies in Buchberger's algorithm, in: Proceedings of the 37th International Conference on Machine Learning, PMLR. pp. 7575–7585. URL: https://proceedings.mlr.press/v119/peifer20a.html.

Rabinowitsch, J.L., 1930. Zum Hilbertschen Nullstellensatz. Mathematische Annalen 102, 520. doi:10.1007/BF01782361.

Rafie-Zinedine, S., 2018. Simplifying Quantum Gravity Calculations. Master's thesis. Lund University. ArXiv:1808.06086.

Ruijl, B., 2026. Symbolica: Modern computer algebra. https://symbolica.io. Version 3.

Shimoyama, T., Yokoyama, K., 1996. Localization and primary decomposition of polynomial ideals. Journal of Symbolic Computation 22, 247–277. doi:10.1006/jsco.1996.0052.

Sommese, A.J., Verschelde, J., Wampler, C.W., 2001. Numerical decomposition of the solution sets of polynomial systems into irreducible components. SIAM Journal on Numerical Analysis 38, 2022–2046. doi:10.1137/S0036142900372549.

Sommese, A.J., Wampler, II, C.W., 2005. The Numerical Solution of Systems of Polynomials Arising in Engineering and Science. World Scientific, Singapore. doi:10.1142/5763.

Sturmfels, B., 2002. Solving Systems of Polynomial Equations. volume 97 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, Providence, RI. doi:10.1090/cbms/097.

Vacca, G.P., Zanusso, O., 2010. Asymptotic safety in einstein gravity and scalar-fermion matter. Physical Review Letters 105, 231601. doi:10.1103/PhysRevLett.105.231601.